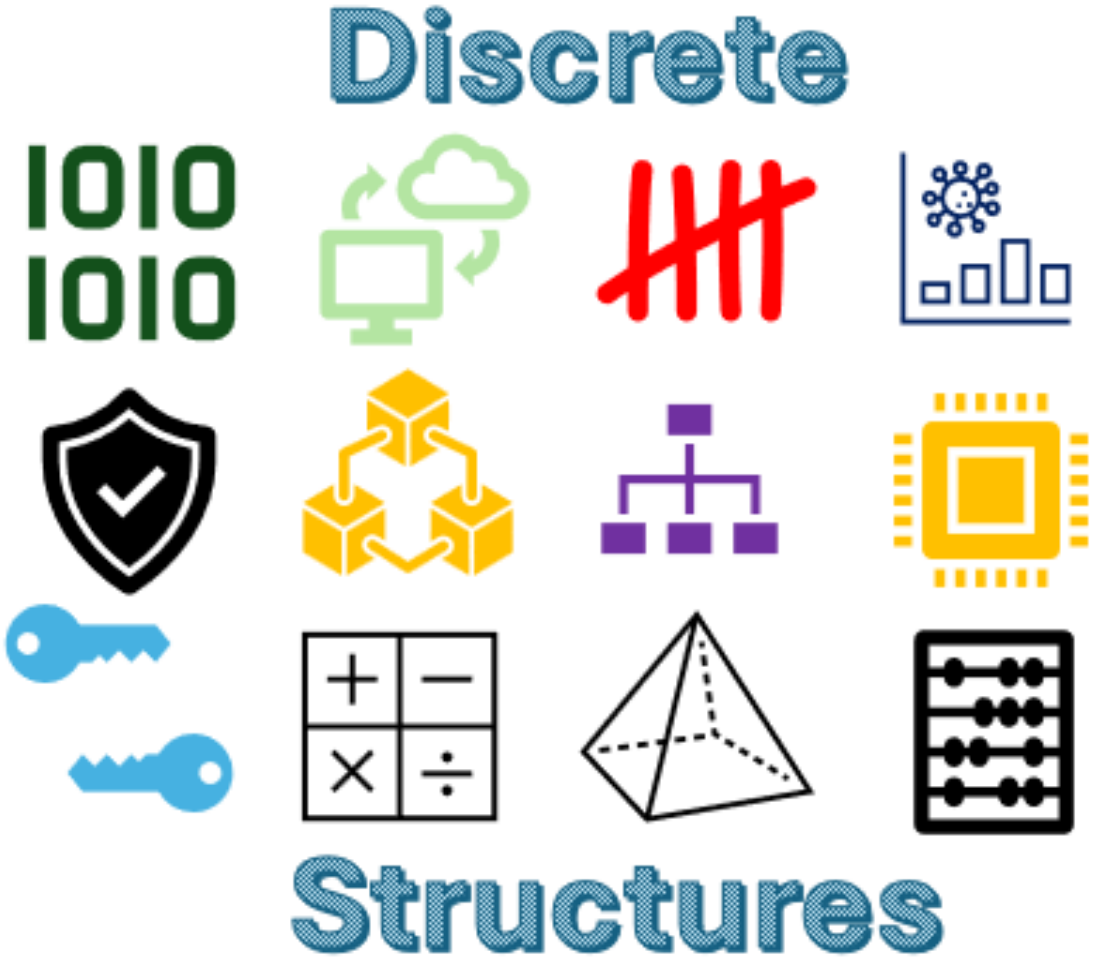
COMP 2310: Discrete Structures
Fall 2026

Chapter 4: Relation and digraph



Cartesian product

If $A_1, A_2, \ldots, A_m$ are nonempty sets, then the ***Cartesian Product*** of them is the set of all ordered m-tuples $(a_1, a_2, \ldots, a_m)$,

where $a_i \in A_i$, i = 1, 2, … m.

Denoted $A_1 \times A_2 \times \ldots \times A_m = \{(a_1, a_2, \ldots, a_m) \mid a_i \in A_i, i = 1, 2, \ldots m\}$

Cartesian product

If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, find $A \times B$

$A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c), (3,a), (3,b), (3,c)\}$

Cartesian product representation using matrices

- For Cartesian Product of two sets, you can use a matrix to find the sets.
- Example: Assume A = {1, 2, 3} and B = {a, b, c}.
- The table below represents A × B.

	a	b	c
1	(1, a)	(1, b)	(1, c)
2	(2, a)	(2, b)	(2, c)
3	(3, a)	(3, b)	(3, c)

Cardinality of cartesian product

- The cardinality of the Cartesian Product equals the product of the cardinality of all of the sets:

$$| A_1 \times A_2 \times \ldots \times A_m | = | A_1 | \cdot | A_2 | \cdot \ldots \cdot | A_m |$$

Subset of cartesian product

- Many of the results of operations on sets produce subsets of the Cartesian Product set
  - Operations: Union, Intersection, difference
- Relational database
  - Each column in a database table can be considered a set
  - Each row is an m-tuple of the elements from each column or set
  - No two rows should be alike

ID	Name	Department
1	Justin	CS
2	Thomas	EE
3	Alice	AG

{(1, Justin, CS), 2, (Thomas, EE), (…),…}

Relation

- A is the set of all students and B is the set of all courses
- A relation R may be defined as the course is required
  - ***Student_A R COMP 2310***

Relation across same set

- Let $A = \{1, 2, 3\}$ and $B = \{r, s\}$. Then $R = \{(1,r), (2, s), (3,r)\}$ is a relation from A to B

Theorem of relations

- Let R be a relation from A to B, and let A_1 and A_2 be subsets of A
  - If $A_1 \subseteq A_2$, then $R(A_1) \subseteq R(A_2)$
  - $R(A_1 \cup A_2) = R(A_1) \cup R(A_2)$
  - $R(A_1 \cap A_2) \subseteq R(A_1) \cap R(A_2)$
- Let R and S be relations from A to B. If R(a) = S(a) for all a in A, then R = S.

Example 16 Let $A = \{1, 2, 3\}$ and $B = \{x, y, z, w, p, q\}$, and consider the relation $R = \{(1, x), (1, z), (2, w), (2, p), (2, q), (3, y)\}$. Let $A_1 = \{1, 2\}$ and $A_2 = \{2, 3\}$. Then $R(A_1) = \{x, z, w, p, q\}$ and $R(A_2) = \{w, p, q, y\}$. Thus $R(A_1) \cup R(A_2) = B$. Since $A_1 \cup A_2 = A$, we see that $R(A_1 \cup A_2) = R(A) = B$, as stated in Theorem 1(b). Also, $R(A_1) \cap R(A_2) = \{w, p, q\} = R(\{2\}) = R(A_1 \cap A_2)$, so in this case equality does hold for the containment in Theorem 1(c). ◆

Matrix of relations

- We can represent a relation between two finite sets with a matrix
- $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 \text{ if } (a_i, b_j) \in R \\ 0 \text{ if } (a_i, b_j) \notin R \end{cases}$$

Matrix of relations

- Using the previous example where A = {1, 2, 3} and B = {a, b, c}. The matrix below represents the relation R = {(1, a), (1, c), (2, c), (3, a), (3, b)}.

	a	b	c
1	1	0	1
2	0	0	1
3	1	1	0

Digraph of relations

- Let R be a relation on A
- We can represent R pictorially as follows
  - Each element of A is a circle called a vertex
  - If a_i is related to a_j, then draw an arrow from the vertex a_i to the vertex a_j
- **In degree** = number of arrows coming into a vertex
- **Out degree** = number of arrows coming out of a vertex

Digraph of relations

The following three representations depict the same relation on A = {1, 2, 3}.

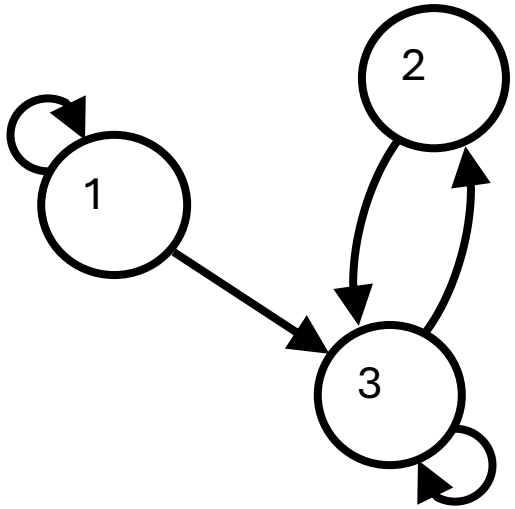
1. Relation using set

R = {(1, 1), (1, 3), (2, 3), (3, 2), (3, 3)}

2. Relation using matrix

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

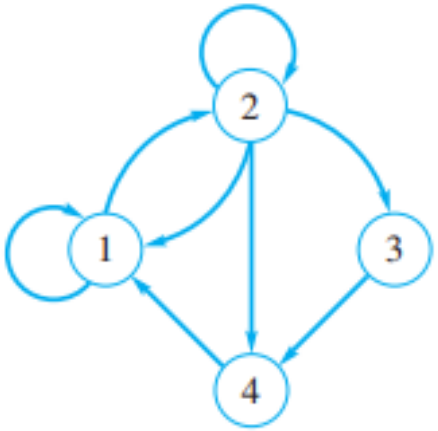
3. Relation using digraph



Node	In Degree	Out Degree
1	1	2
2	1	1
3	3	2

Digraph of relations: example

A = {1, 2, 3, 4}
R = {(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4), (4, 1)}.



$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

- Since **M** is 3 × 4, we let $A = \{a1, a2, a3\}$ and $B = \{b1, b2, b3, b4\}$
- Then $(ai, bj) \in R$ if and only if $mij = 1$.
- Therefore, $R = \{(a1, b1), (a1, b4), (a2, b2), (a2, b3), (a3, b1), (a3, b3)\}$.

Digraph of relations: example

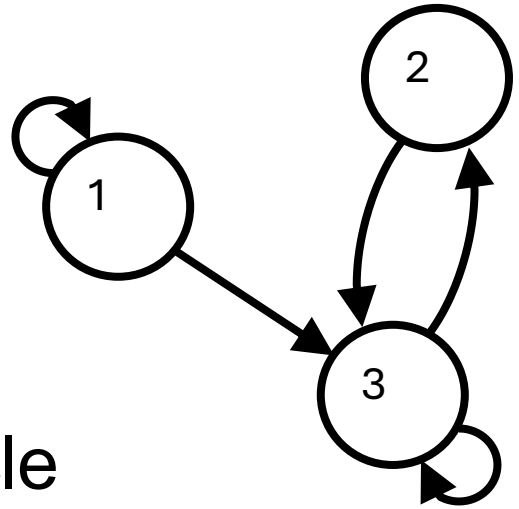
- Let $A = \{a, b, c, d\}$, and let R be the relation on A that has the matrix

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

- Construct the digraph of R, and list in-degrees and out-degrees of all vertices.

Paths in digraph

- A walk from one node to the next, following the arrows is a Path
  - Vertex1, Edge1, Vertex2, Edge2, Vertex3, Edge3, Vertex4
  - A vertex may occur multiple times in a path; an edge may occur only once
- Paths start and end at a vertex
- The number of edges in the path is its Length
  - The example to the right as a length 3
- A path that starts and ends at the same node is a cycle

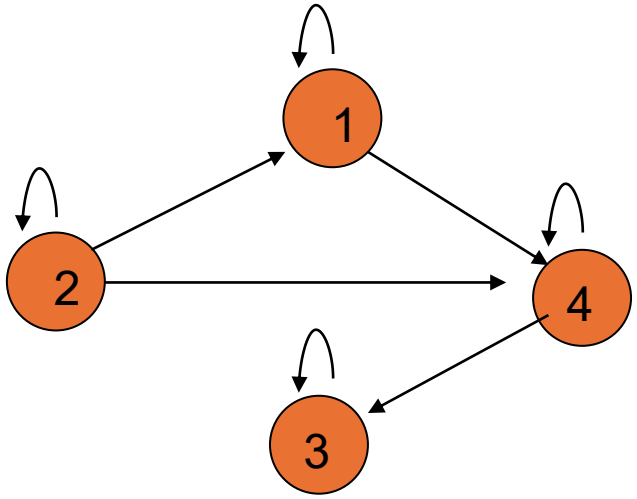


Properties of relations

- Recall
  - If $Q \subseteq A \times A$, then Q is a relation on A
- Relations on a set may have three properties
  - Reflexive
  - Symmetric
  - Transitive

Properties of relations: reflexive

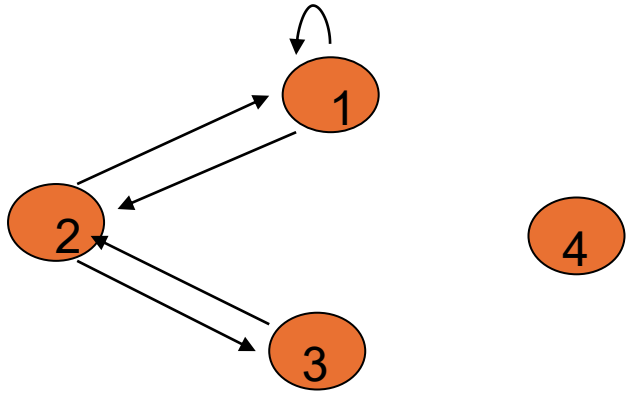
- A relation Q on set A, is reflexive if for all $a \in A$, then $(a, a) \in Q$
- Example: Let A={1, 2, 3, 4}
  - Q={ (1,1), (1,4), (2,1), (2,2), (2,4), (3,3), (4,3), (4,4) }
- ***Must have self pairs for all elements***



- A relation Q on set A, is **irreflexive** if no element relates to itself

Properties of relations: symmetric

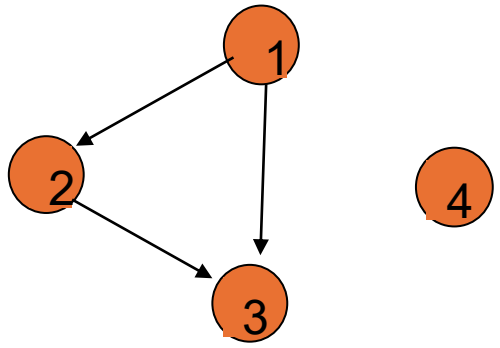
- A relation Q on set *A*, is symmetric
  - If $(a, b) \in Q$ then (b, a) is also $\in Q$
- $A = \{1, 2, 3\}$
  - $Q = \{ (1, 1), (1, 2), (2, 1), (2, 3), (3, 2) \}$



- A relation *Q* on set *A*, is **asymmetric** if whenever $(a, b) \in Q$, $(b, a) \notin Q$
  - If the reverse pair is not available

Properties of relations: transitive

- A relation Q on set A, is transitive if $(a, b) \in Q$ and $(b, c) \in Q$ then $(a, c) \in Q$
- Example: Let A={ 1, 2, 3 } R={ (1, 2), (2, 3), (1, 3) }
- If an element a relates to b (written as (a,b)∈Q) and that element b relates to c (written as (b,c)∈Q, then the **first element** a should also **directly** relate to c (i.e., (a,c)∈Q).
- If your friend A is related to B and you as C are also related to B



Equivalence relation

- A relation *Q* on a set *A* is an equivalence relation if the relation is
  - Reflexive
  - Symmetric
  - Transitive

Equivalence relation: example

- Determine whether the relation R is an equivalence relation

A = {a, b, c, d},
R = {(a,a), (b,a), (b, b), (c, c), (d, d), (d, c)}

The relation R is not equivalence